

Retail Superstore Sales Performance Analysis

SQL + Power BI

**Presented by
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Project Overview

TOOLS:

SQL
Power BI
Excel

----- DATASET:

Global Superstore

----- ROLE:

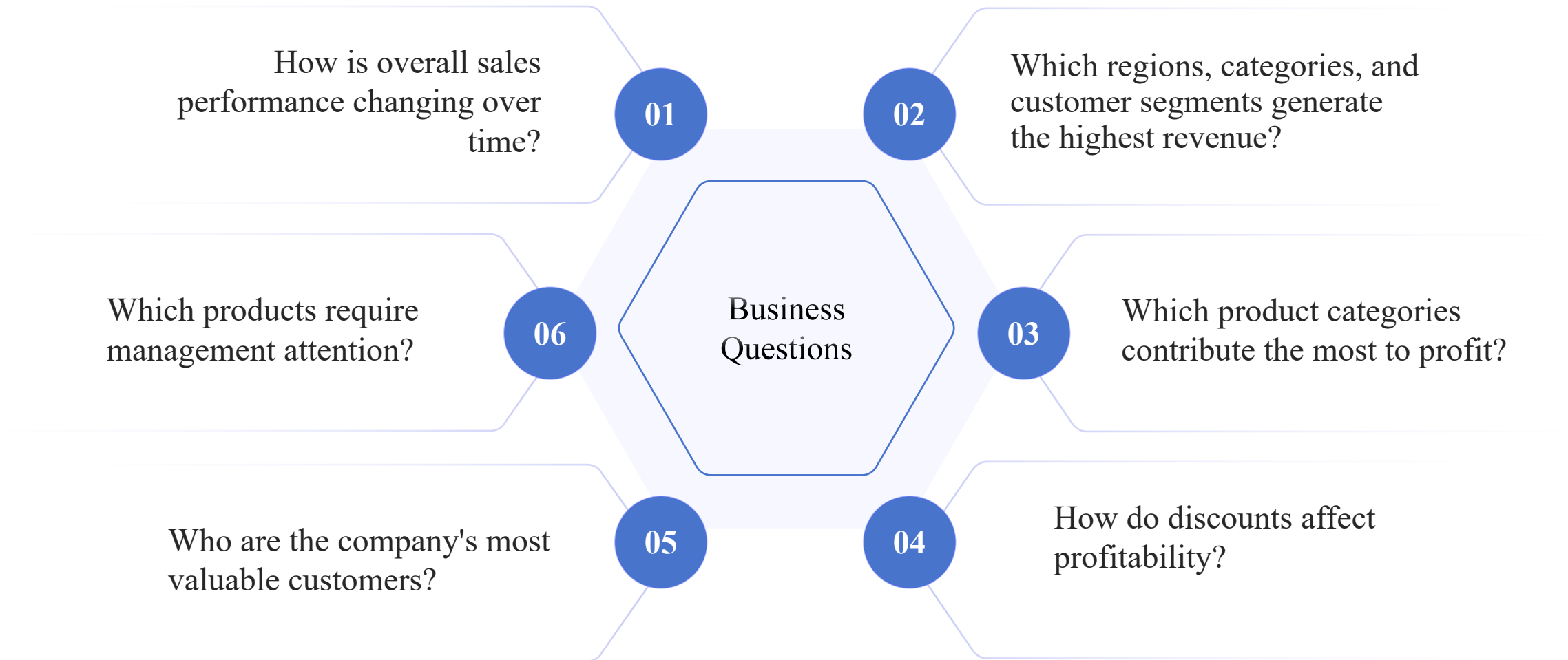
Data Analyst

Global Superstore wanted to understand how its sales performance, profitability, customer behavior, and product performance changed over time.

Using SQL for data analysis and Power BI for visualization, I analyzed transactional data to identify trends, uncover business opportunities, and provide actionable recommendations that support better decision-making.

Objective

Transform raw sales data into meaningful business insights through data cleaning, SQL analysis, interactive dashboards, and data storytelling.



This analysis was conducted to answer the above business questions

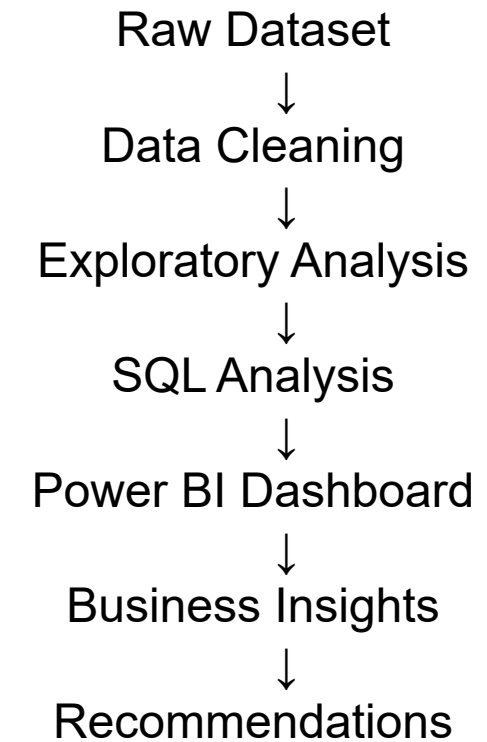
Before beginning the analysis, the dataset was explored to understand its structure, data types, and overall quality.

The preparation process included:

1. Reviewing all variables and their business meaning
2. Identifying missing values and inconsistencies
3. Cleaning and standardizing the dataset
4. Preparing the data for SQL analysis and Power BI visualization

This ensured the analysis was based on reliable and consistent data.

Data Understanding and Preparation



SQL Analysis

SQL was used to explore the dataset, answer key business questions, and prepare insights for visualization.

The analysis focused on:

Sales performance

01

Product performance

04

Profitability

02

Discount impact

05

Customer behavior

03

Time-based trends

06

The resulting insights formed the foundation of the interactive Power BI dashboard.

EXECUTIVE DASHBOARD

EXECUTIVE SUMMARY

Total Sales

\$12.64M

Total Profit

\$1.47M

Total Order Quantity

178K

Total Discount

\$7.33K

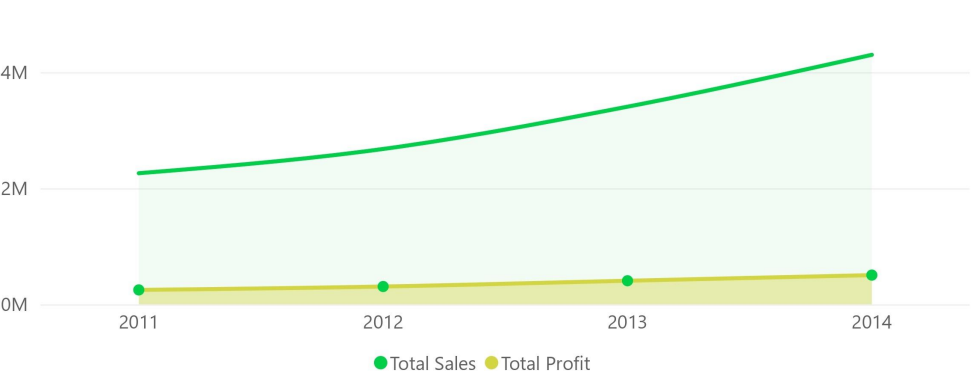
Profit Margin

11.61%

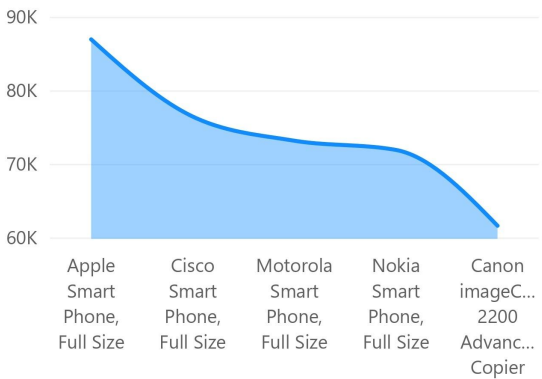
Total No. of Customers

795

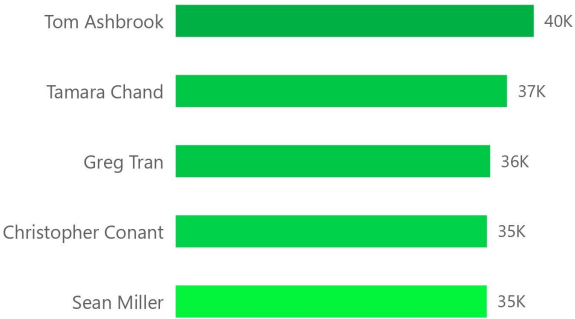
Year-on-Year Sales & Profit



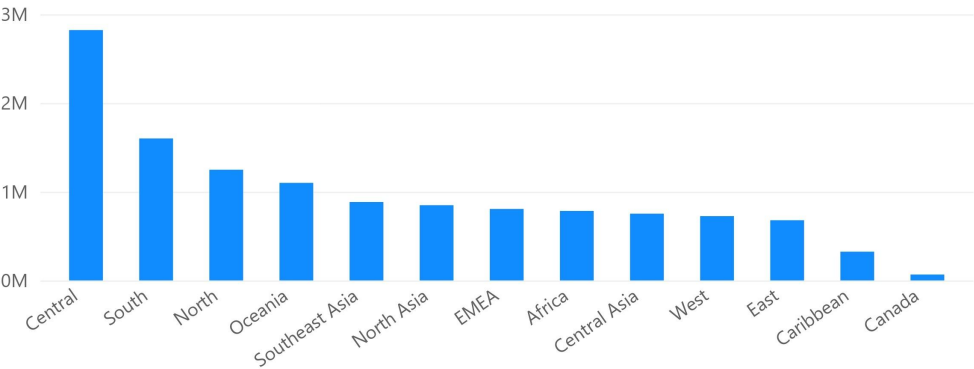
Top 5 Products



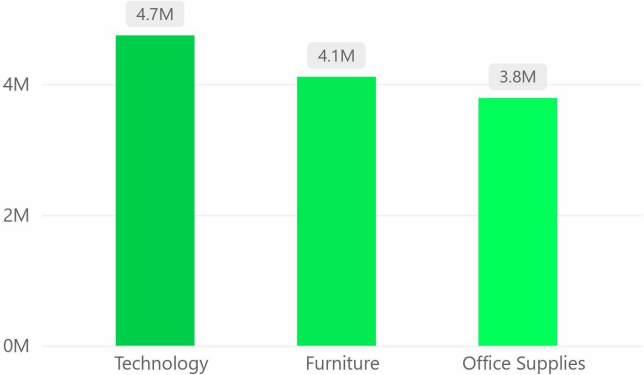
Top 5 Customers



Sales by Region



Sales by Category



Year, Month

- ☐ Select all
- ☒ 2011
- ☒ 2012
- ☒ 2013
- ☐ January

Executive Insight

Business Question 1: How is the business performing overall?

Finding

Sales increased steadily between 2011 and 2014, with 2014 recording the strongest overall performance. The business also exhibits a clear seasonal trend, as Q4, particularly November and December, consistently generated the highest sales, with November 2014 reaching a peak of 555,312.

Business Impact

The recurring Q4 sales surge presents an opportunity to maximize revenue through proactive inventory planning, marketing campaigns, and staffing.

Business Question 2: Which regions, categories, and customer segments generate the highest revenue?

Finding

The Central region generated the highest revenue (2.82M), while the United States emerged as the company's strongest market (2.30M). Across customer segments, the Consumer segment contributed the largest share of revenue.

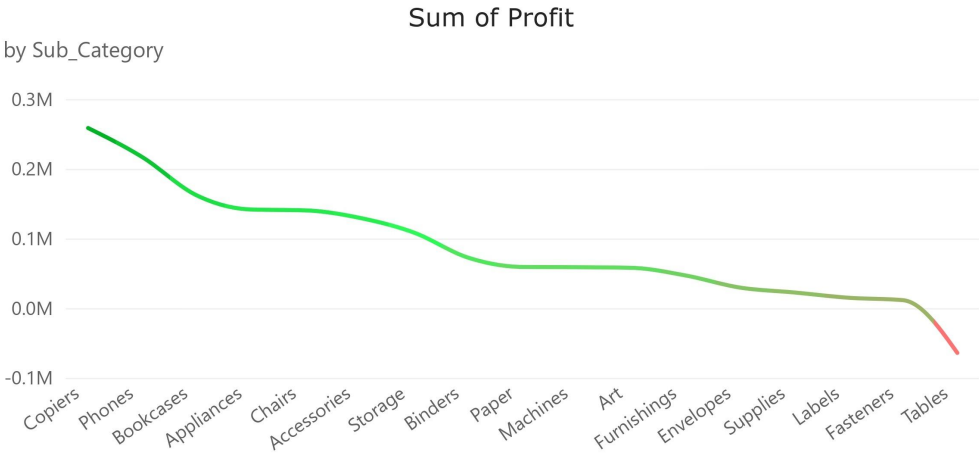
Business Impact

The Central region, United States, and Consumer segment represent the company's primary revenue drivers and should remain strategic priorities for investment and customer retention.

PROFITABILITY DASHBOARD

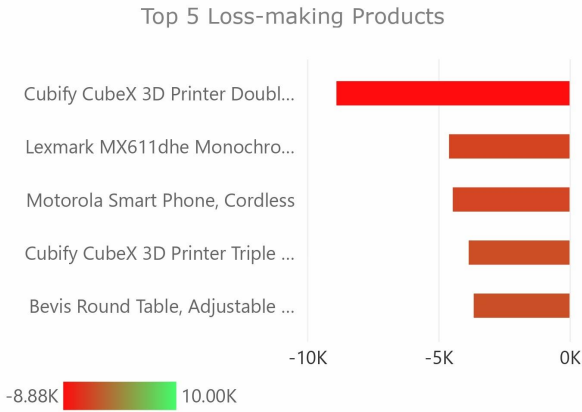
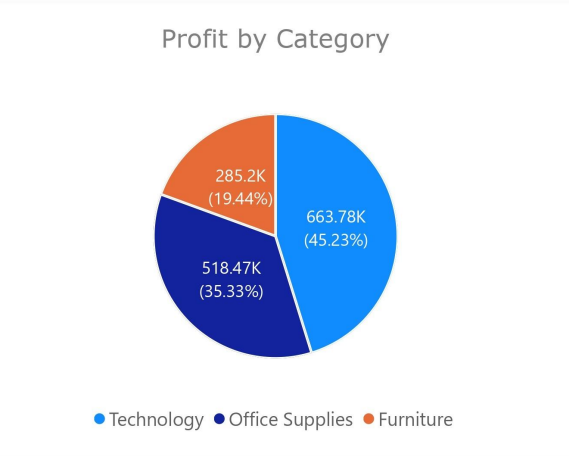
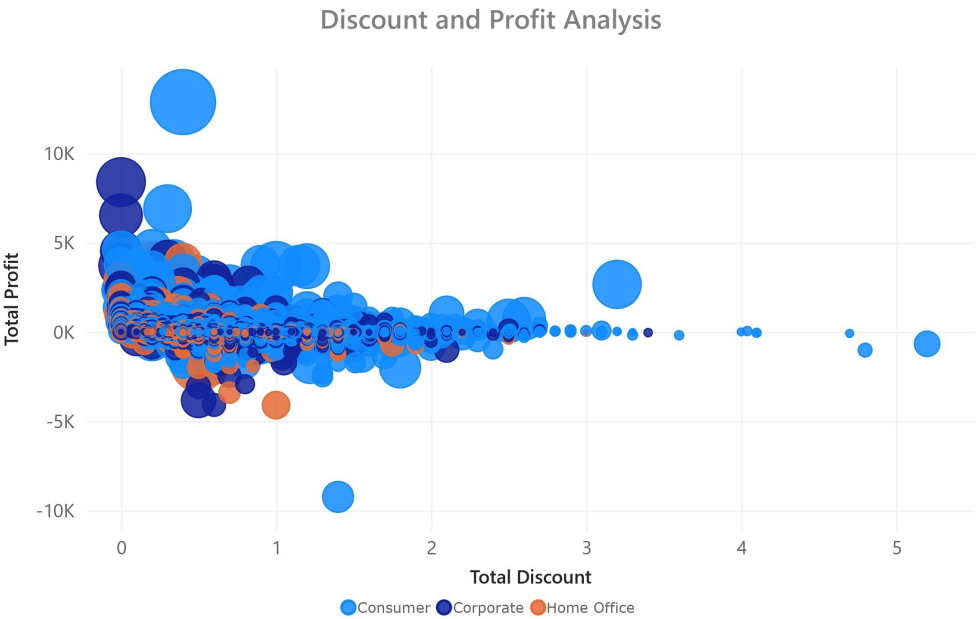
PROFITABILITY INSIGHT

1.47M
Total Profit



Profit by Region

Region	2011	2012	2013	2014
Africa	10,944.11	11,908.94	26,687.11	39,331.47
Canada	1,807.08	4,887.84	5,129.46	5,993.01
Caribbean	4,358.91	8,706.26	8,973.54	12,532.62
Central	52,678.47	63,617.31	97,384.63	97,723.57
Central Asia	22,846.45	28,977.08	33,109.18	47,547.48
East	17,059.61	21,091.01	20,176.81	33,195.35
EMEA	5,280.03	5,419.58	10,598.04	22,600.32
North	35,866.23	50,905.94	51,167.43	56,658.35
North Asia	35,513.37	28,020.32	49,274.29	52,770.44
Oceania	21,428.76	29,675.32	37,553.45	31,431.58
South	17,849.10	30,975.36	39,755.14	51,776.16
Southeast Asia	3,243.00	2,738.13	3,166.20	8,705.00
West	20,065.69	20,492.19	23,959.94	43,900.63



Profitability Insight

Business Question 3: Which product categories contribute the most to profit?

Finding

The Technology category generated the highest profit, making it the company's strongest contributor to overall profitability.

Business Impact

Technology products should remain a strategic focus while management reviews lower-performing categories to identify opportunities for margin improvement.

Discount Insight

Business Question 4: How do discounts affect profitability?

Finding

The analysis shows a negative relationship between discounts and profit. As discount levels increase, profitability generally declines, with heavily discounted transactions more likely to generate low or negative profits.

Business Impact

Discounting should be carefully controlled. Establishing discount thresholds and regularly monitoring their impact can help improve overall profit margins.

CUSTOMER DASHBOARD

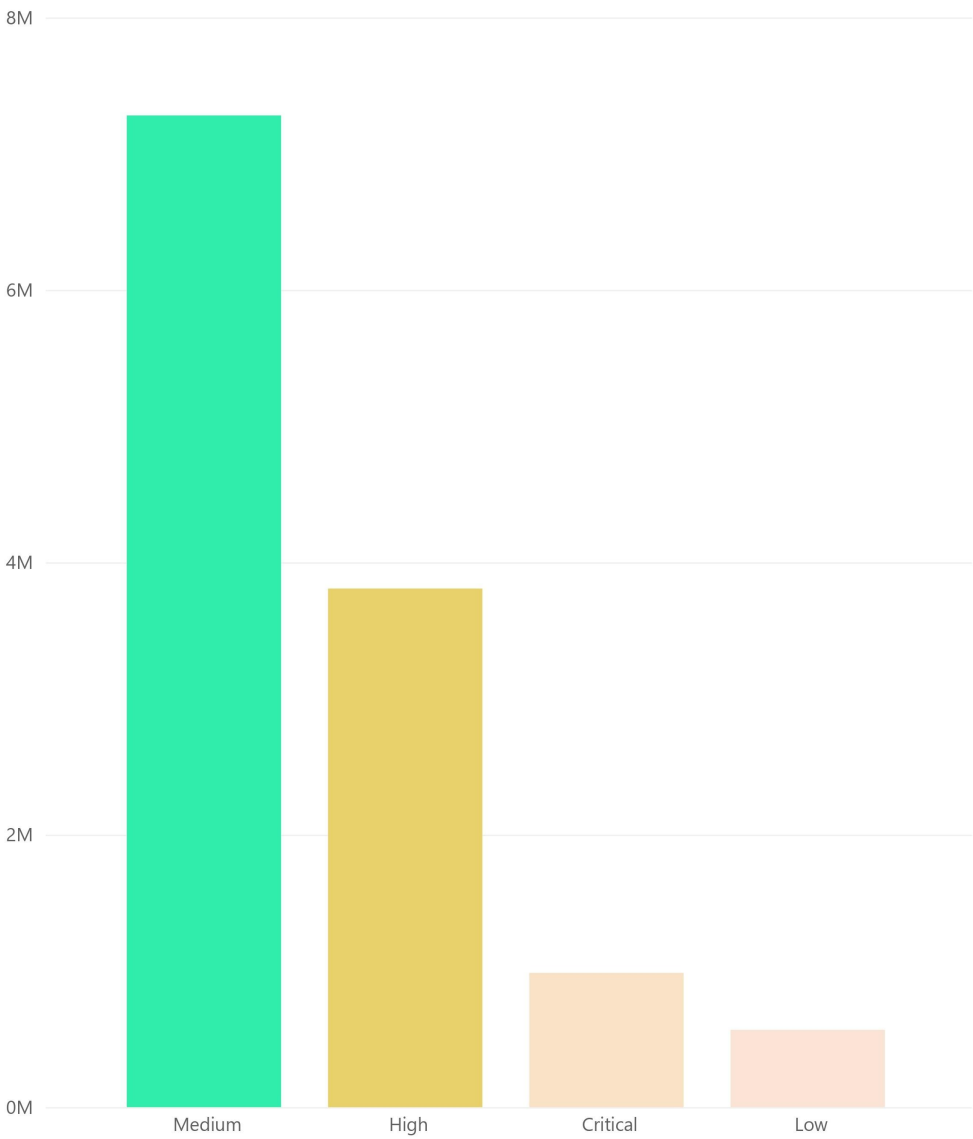
CUSTOMER INSIGHT

795
Total Customer

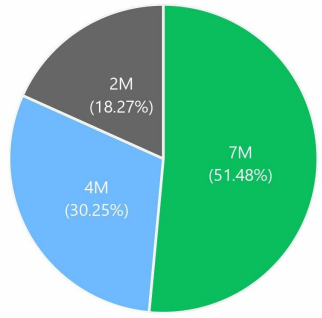
Top 10 Customers



Customers' Order priority

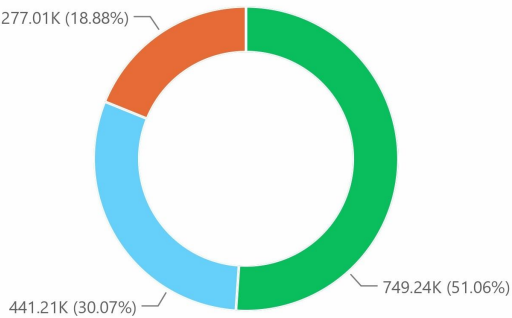


Sales by Customer Segment



● Consumer ● Corporate ● Home Office

Profit by Segment



● Consumer ● Corporate ● Home Office



Customer Insight

Business Question 5: Who are the Company's most valuable customers

The company serves 795 customers, with revenue concentrated among a relatively small group of high-value accounts. Tom Ashbrook generated the highest sales (40.5K), followed by Tamara Chand, Greg Tran, Christopher Conant, and Sean Miller. While these customers represent only a small fraction of the customer base, they contribute significantly to overall revenue.

Business Impact

Retaining high-value customers should be a strategic priority. Strengthening relationships through loyalty programs, personalized engagement, and dedicated account management can help protect revenue while reducing customer churn.

Products Requiring Immediate Review

Business Question 6: Which products require management attention?

Several premium products, including the Cubify CubeX 3D Printer Double Head, Lexmark MX611dhe Laser Printer, and Motorola Smart Phone, consistently generated losses despite their high market value. These losses may stem from excessive discounting, high procurement costs, warranty expenses, or poor pricing strategies.

Business Impact

Management should review these products to determine whether losses are driven by excessive discounting, high procurement costs, warranty expenses, or pricing strategies. If profitability cannot be improved, discontinuing or repositioning these products should be considered.

TIME ANALYSIS TREND

TIME ANALYSIS

Select all

2011

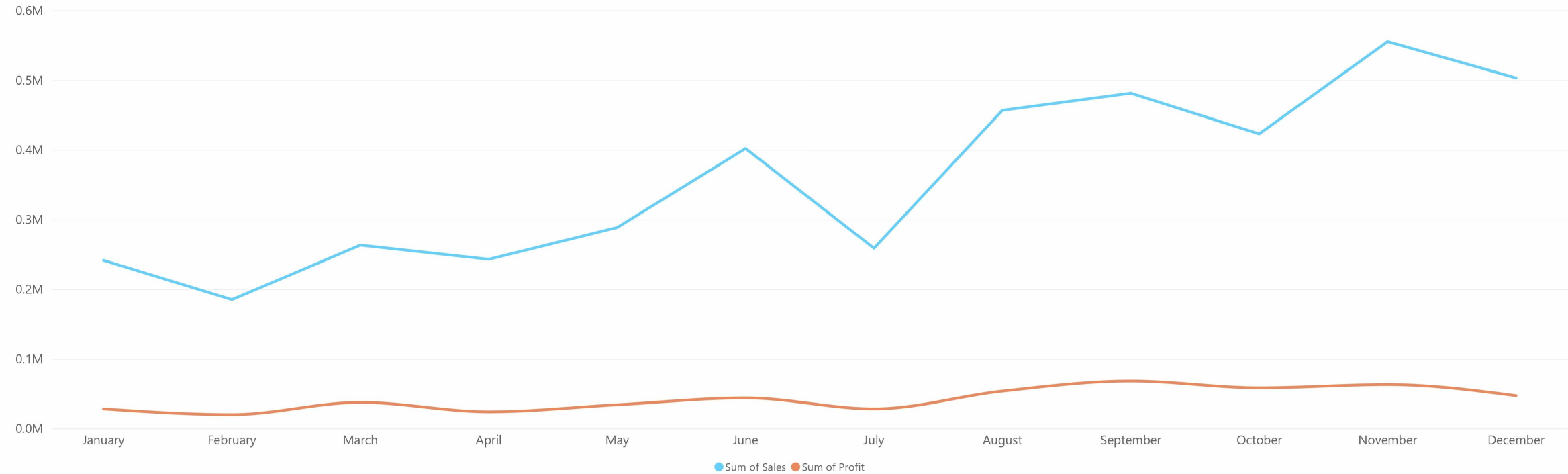
2012

2013

2014



Monthly Sales & Profit Trend



Quarterly Sales & Profit Trend



RECOMMENDATIONS

Based on the findings from this analysis, the following actions are recommended to improve sales performance and profitability.

Optimize Discount Strategy

Review and standardize discount policies to prevent excessive discounting that erodes profit margins.

Capitalize on Peak Sales Periods

Increase inventory, marketing campaigns, and operational capacity ahead of Q4 to maximize revenue during the strongest sales season.

01

02

03

Improve Performance in Low-Margin Markets

Investigate regions and countries with strong sales but weak profitability to identify pricing, cost, or operational inefficiencies.

Review Loss-Making Products

Assess the pricing, procurement costs, and profitability of consistently loss-making products. Consider repricing, renegotiating supplier costs, or discontinuing products that remain unprofitable.

Retain High-Value Customers

Develop loyalty initiatives and personalized engagement strategies for top customers to strengthen long-term customer retention.

04

05

06

Build on High-Performing Markets

Leverage the strategies driving success in the Central region and the United States to improve performance in underperforming markets.

THANK YOU

I welcome feedback and opportunities to collaborate on data-driven projects.



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Appendix: Sample SQL Queries

The following queries were used to answer key business questions and prepare insights for visualization in Power BI.

Sales by Segment

Query 1

```
1 • SELECT
2   segment,
3   SUM(sales) AS total_sales
4 FROM superstore_clean
5 GROUP BY segment
6 ORDER BY total_sales DESC;
```

Result Grid

segment	total_sales
Consumer	6508141
Corporate	3824808
Home Office	2309956

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Discount vs Profit Analysis

Query 1

```
1 • SELECT
2   discount,
3   COUNT(*) AS no_of_orders,
4   ROUND(SUM(sales), 2) AS total_sales,
5   ROUND(SUM(profit), 2) AS total_profit,
6   ROUND(AVG(profit), 2) AS avg_profit
7 FROM superstore_clean
8 GROUP BY discount
9 ORDER BY discount;
```

Result Grid

discount	no_of_orders	total_sales	total_profit	avg_profit
0.00	29470	7254129	1828671.85	62.05
0.07	150	121620	21148.5	140.99
0.10	4068	1579618	259064.18	63.68
0.15	541	306037	27375.9	50.6
0.17	735	247404	28163.07	38.32
0.20	5039	1220068	117120.59	23.24
0.25	198	88063	800.59	4.04
0.27	388	89351	-1675.08	-4.32
0.30	340	188913	-19685.85	-57.9
0.32	27	14491	-2391.14	-88.56
0.35	122	95704	-14169.65	-116.14
0.37	74	31615	-5806.2	-78.46

Result 5

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Monthly sales & profit Trend

Query 1 x

```
1 SELECT
2     DATE_FORMAT(Order_Date, '%Y-%m') AS month,
3     SUM(sales) AS total_sales,
4     SUM(profit) AS total_profit
5 FROM superstore_clean
6 GROUP BY DATE_FORMAT(order_date, '%Y-%m')
7 ORDER BY DATE_FORMAT(order_date, '%Y-%m') DESC;
```

Result Grid

month	total_sales	total_profit
2014-12	503154	46916.52068
2014-11	555312	62856.5879000001
2014-10	422785	58209.834760000034
2014-09	481186	67979.45109999998
2014-08	456633	53542.894959999954
2014-07	258718	28035.87258000002
2014-06	401843	43778.60280000004
2014-05	288404	33953.557740000055
2014-04	242781	23782.301200000013
2014-03	263110	37357.260519999974

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Top Customers

Query 1 x

```
1 SELECT
2     customer_name,
3     ROUND(SUM(sales), 2) AS total_sales
4 FROM superstore_clean
5 GROUP BY customer_name
6 ORDER BY total_sales DESC
7 LIMIT 10;
```

Result Grid

customer_name	total_sales
Tom Ashbrook	40489
Tamara Chand	37453
Greg Tran	35552
Christopher Conant	35187
Sean Miller	35170
Bart Watters	32315
Natalie Fritzier	31778
Fred Hopkins	30404
Jane Waco	30288
Hunter Lopez	30246

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